MANY ISOLATED REAL ZEROS OF SUMS OF SQUARES

Abstract. Let $\widetilde{\#}(2k,\ell)$ be the largest number of isolated real zeros of a sum of squares of real polynomials of degree at most k in ℓ variables. We construct an explicit quartic sum of nine squares in ten variables with exactly 1152 isolated real zeros, exceeding 2^{10}. This refutes the Ottaviani–Shapiro conjecture $\widetilde{\#}(2k,\ell)=k^\ell$. We also give a uniform construction proving $\widetilde{\#}(2k,\ell)\geq((\ell-1)(k-1)/4)k^{\ell-1}$ for every even $k\geq 2$ and $\ell\geq 3$; no exact extremal formula or matching upper bound is claimed.

1. The conjecture and the counterexample

For integers $k,\ell\geq 1$, let $\widetilde{\#}(2k,\ell)$ denote the largest possible number of isolated real zeros of a polynomial in ℓ variables that is a finite sum of squares of real polynomials of degree at most k. Ottaviani and Shapiro conjectured that

$$\widetilde{\#}(2k,\ell)=k^\ell \tag{1}$$

in Shapiro's problem collection [1, Problem 3 and Conjecture 4].

Theorem 1.1. *There is a quartic sum of nine squares in ten real variables having exactly* 1152 *isolated real zeros. Consequently*

$$\widetilde{\#}(4,10)\geq 1152>1024=2^{10},$$

so (1) *is false.*

Proof. Use the variables $t,x_0,x_1,\dots,x_8$ and define

$$q_0=x_0^2+t(t-8),\qquad q_i=x_i^2-(t-i+1)(t-i)\quad(1\leq i\leq 8). \tag{2}$$

Set

$$P=q_0^2+q_1^2+\cdots+q_8^2.$$

Then P is nonnegative, is a sum of nine squares of quadratic polynomials, and has degree four. Over the reals, $P=0$ if and only if every q_i is zero, equivalently

$$x_0^2=t(8-t),\qquad x_i^2=(t-i+1)(t-i)\quad(1\leq i\leq 8). \tag{3}$$

The first equation forces $0\leq t\leq 8$. For $i\geq 1$, the right-hand side is nonnegative exactly when $t\leq i-1$ or $t\geq i$, so the ith equation excludes $(i-1,i)$. Therefore $t\in\{0,1,\dots,8\}$.

Conversely, every one of these nine values is feasible. At $t=j$, exactly two of the nine radicands in (3) vanish: x_0,x_1 when $j=0$; x_j,x_{j+1} when $1\leq j\leq 7$; and x_0,x_8 when $j=8$. The other seven radicands are strictly positive and admit two independent sign choices each. Thus every t-slice contains $2^7=128$ points, and

$$\#Z_{\mathbb{R}}(P)=9\cdot 2^7=1152.$$

The real zero set is finite, hence all of its points are isolated. □

2. Algebraic structure of the base example

For $j\in\{0,\dots,8\}$, put

$$a_0(j)=j(8-j),\qquad a_i(j)=(j-i+1)(j-i)\quad(1\leq i\leq 8),$$

and define

$$K_j=\big(t-j,\ x_i \text{ for } a_i(j)=0,\ x_i^2-a_i(j) \text{ for } a_i(j)>0\big).$$

Proposition 2.1. *If $I = (q_0, \dots, q_8) \subset \mathbb{R}[t, x_0, \dots, x_8]$, then*

$$\sqrt[\mathbb{R}]{I} = \bigcap_{j=0}^{8} K_j.$$

Each K_j is real radical, and its real variety consists of 2^7 points.

Proof. Every positive quadratic $x_i^2 - a_i(j)$ splits into two distinct real linear factors. Since the seven factorizations involve distinct variables, K_j is the intersection of 2^7 real maximal ideals. The sign analysis in the proof of Theorem 1.1 shows that these nine fibers are exactly the real common-zero set of I. The real Nullstellensatz gives the displayed decomposition. □

The complex common-zero locus is not zero-dimensional. Its coordinate ring is

$$A = \mathbb{C}[t, x_0, \dots, x_8]/\big(x_0^2 - t(8-t),\ x_i^2 - (t-i+1)(t-i) : 1 \le i \le 8\big).$$

Successive division by the nine monic quadratics, each in its own variable, gives the explicit $\mathbb{C}[t]$-basis

$$x_0^{e_0} x_1^{e_1} \cdots x_8^{e_8}, \qquad e_i \in \{0, 1\}.$$

Hence A is finite free of rank $2^9 = 512$ over $\mathbb{C}[t]$. Equivalently, the nine quadratics form a regular sequence and define an unmixed one-dimensional complete intersection. Therefore every irreducible component of the complex common-zero locus is one-dimensional; there is no hidden vertical zero-dimensional component. The isolated real points lie on positive-dimensional complex components, which is why a naive zero-dimensional Bezout count does not apply.

3. A general even-degree family

The same selector construction yields a uniform lower-bound family.

Theorem 3.1. *For every even $k \ge 2$ and every $\ell \ge 3$,*

$$\widetilde{\#}(2k, \ell) \ge \frac{(\ell-1)(k-1)}{4}\, k^{\ell-1}. \tag{4}$$

In particular, the conjectured value k^ℓ is exceeded whenever $(\ell-1)(k-1) > 4k$.

Proof. Write $k = 2d$, put $m = \ell - 1$ and $N = md$, and use variables $t, x_1, \dots, x_m$. Define degree-k sign polynomials

$$A_1(t) = -(t-d)(t-N) \prod_{r=1}^{d-1} (t-r)^2,$$
$$A_i(t) = (t-(i-1)d)(t-id) \prod_{r=(i-1)d+1}^{id-1} (t-r)^2 \quad (2 \le i \le m),$$

where an empty product equals one. Their simultaneous nonnegative locus is exactly $\{1, 2, \dots, N\}$. Indeed, A_1 excludes both exterior regions and every nonintegral point of the first block, while A_i is negative at every nonintegral point of its own block $((i-1)d, id)$ for $i \ge 2$. Every retained integer makes all A_i nonnegative.

Among the retained integers, $m(d-1)$ are internal block points at which exactly one A_i vanishes, and m are cyclic block-boundary points at which exactly two A_i vanish. Define

$$F_d(x) = \prod_{r=1}^{d} (x-r)^2$$

and

$$\eta_d = \min\{F_d(r + \sigma/3) : 1 \le r \le d,\ \sigma \in \{-1, 1\}\} > 0.$$

Let

$$M = \max\{A_i(j) : 1 \le i \le m,\ 1 \le j \le N\} \ge 0, \qquad \varepsilon = \frac{\eta_d}{2(1+M)}.$$

The weak inequality $M \geq 0$ includes the boundary case $(k, \ell) = (2, 3)$, where $M = 0$. Put

$$q_i(x_i, t) = F_d(x_i) - \varepsilon A_i(t), \qquad P_{k,\ell} = \sum_{i=1}^{m} q_i(x_i, t)^2.$$

Every q_i has degree at most k. If $t \notin \{1, \ldots, N\}$, some $A_i(t) < 0$, so the corresponding equation $q_i = 0$ has no real solution.

At an allowed integer, if $A_i = 0$, then $q_i = F_d$ has exactly $d = k/2$ distinct real roots. If $A_i > 0$, put $c = \varepsilon A_i$. Then $0 < c < \eta_d$. On each of the d disjoint intervals $(r - 1/3, r + 1/3)$, the polynomial $F_d - c$ is positive at both endpoints and negative at r. Hence it has at least two roots in each interval. Its degree is $2d = k$, so these are exactly k distinct real roots.

A one-active integer therefore contributes $(k/2)k^{m-1}$ points, and a two-active integer contributes $(k/2)^2 k^{m-2}$. Summing gives

$$\begin{aligned} \#Z_{\mathbb{R}}(P_{k,\ell}) &= m(d-1)\frac{k}{2}k^{m-1} + m\left(\frac{k}{2}\right)^2 k^{m-2} \\ &= \frac{m(k-1)}{4}k^m, \end{aligned}$$

which is (4). The real zero set is finite, so all counted points are isolated. □

The family refutes (1), for example, when $k = 2$ and $\ell \geq 10$, when $k = 4$ and $\ell \geq 7$, and for every even $k \geq 6$ and $\ell \geq 6$.

Corollary 3.2. *For every $k \geq 2$ and $\ell \geq 3$, set $e = 2\lfloor k/2 \rfloor$. Then*

$$\widetilde{\#}(2k, \ell) \geq \frac{(\ell - 1)(e - 1)}{4} e^{\ell - 1}.$$

Proof. Apply Theorem 3.1 with summand degree $e \leq k$. For odd $k \geq 3$, this uses $e = k - 1$. No claim is made for $k = 1$. □

If two admissible sums of squares in disjoint variable sets have finite real zero sets of sizes R_1 and R_2, their sum has Cartesian-product zero set of size $R_1 R_2$. Thus strict block violations can be amplified through independent blocks.

4. Scope

The construction refutes the universal equality (1); it does not determine the exact extremal function. In particular, it does not prove $\widetilde{\#}(4, 10) = 1152$, provide a matching upper bound for (4), or classify extremizers.

References

[1] B. Shapiro, *Problems around polynomials: the good, the bad and the ugly*, Arnold Math. J. **1** (2015), 91–99. doi:10.1007/s40598-015-0008-4. arXiv:1503.05295.